

# Solution of the tutorial on parametric estimation of rational spectra - TSIA202b

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## Exercise 1

Zero the Lagrangian of the cost function  $J(p_X) = -H_X = \int_{\mathbb{R}} p_X(x) \ln(p_X(x))$  with constraints  $-p_X(x) \leq 0$   $\forall x \in \mathbb{R}$  (actually this constraint is inactive and can be discarded),  $\int_{\mathbb{R}} p_X(x) dx = 1$ ,  $\int_{\mathbb{R}} x p_X(x) dx = \mu_X$ , and  $\int_{\mathbb{R}} x^2 p_X(x) dx = \mu_X^2 + \sigma_X^2$ .

## Exercise 2

We have  $\sigma_f^2 = \mathbb{E} \{ |Z_t^f|^2 \} = \begin{bmatrix} 1 & -\theta_f^H \end{bmatrix} R_{p+1} \begin{bmatrix} 1 \\ -\theta_f \end{bmatrix}$  where  $R_{p+1} = \begin{bmatrix} r_{YY}(0) & r_{YY}(-1) & \dots & r_{YY}(-p) \\ r_{YY}(1) & r_{YY}(0) & & \vdots \\ \vdots & & \ddots & r_{YY}(-1) \\ r_{YY}(p) & \dots & & r_{YY}(0) \end{bmatrix}$ .

The minimization of  $\sigma_f^2$  w.r.t.  $\theta_f$  yields equation (1), where  $\sigma_p^2$  is simply defined as  $r_{YY}(0) - \sum_{k=1}^p a_k r_{YY}(-k)$ . Then substituting equation (1) into the expression of  $\sigma_f^2$  yields  $\sigma_f^2 = \sigma_p^2$ .

## Exercise 3

We have  $\sigma_b^2 = \mathbb{E} \{ |Z_t^b|^2 \} = \begin{bmatrix} 1 & -\theta_b^\top \end{bmatrix} R_{p+1} \begin{bmatrix} 1 \\ -\theta_b \end{bmatrix}$  where  $R_{p+1}$  is the same matrix as in Exercise 2.

This expression is formally the same as the expression of  $\sigma_f^2$ , with  $\theta_f$  replaced by  $\bar{\theta}_b$ . Therefore the minimum is the same:  $\sigma_b^2 = \sigma_f^2$ , and it is reached at  $\theta_b = \bar{\theta}_f$ .

## Exercise 4

$$1. \ R_{2q+1} \begin{bmatrix} -d_p \\ \vdots \\ -d_1 \\ 1 \\ -c_1 \\ \vdots \\ -c_p \end{bmatrix} = \begin{bmatrix} 0 \\ \vdots \\ 0 \\ \sigma_s^2 \\ 0 \\ \vdots \\ 0 \end{bmatrix}$$

2. This can be proved by conjugating the previous linear system, reversing its rows, and reversing both the columns of the matrix and the coefficients of the solution vector, leading to the same linear system, with  $c_k$  replaced by  $\bar{d}_k$  and  $d_k$  replaced by  $\bar{c}_k$ .
3.  $\sigma_s^2$  is the unique minimum of  $J(\theta) = \theta^H R_{2p+1} \theta$  and this minimum is reached for

$$\theta_s = \begin{bmatrix} -\bar{c}_p \\ \vdots \\ -\bar{c}_1 \\ 1 \\ -c_1 \\ \vdots \\ -c_p \end{bmatrix}. \text{ However, } \sigma_f^2 = \theta_f^H R_{2p+1} \theta_f \text{ for } \theta_f = \begin{bmatrix} 0 \\ \vdots \\ 0 \\ 1 \\ -a_1 \\ \vdots \\ -a_p \end{bmatrix}, \sigma_b^2 = \theta_b^H R_{2p+1} \theta_b \text{ for } \theta_b = \begin{bmatrix} -\bar{a}_p \\ \vdots \\ -\bar{a}_1 \\ 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix}.$$

Since  $\theta_f \neq \theta_s$  and  $\theta_b \neq \theta_s$ , we conclude that  $\sigma_f^2 = \sigma_b^2 > \sigma_s^2$ .

## Exercise 5

Decompose the ARMA system as  $Y_t = \frac{1}{A(z)}Z_t$  and  $X_t = B(z)Y_t$ . Then  $\{Y_t\}$  is an AR( $p$ ) process. To find  $\{r_{YY}(k)\}$  from  $\{\sigma_Z^2, a_1 \dots a_p\}$ , write the Yule-Walker equations as:

$$\begin{bmatrix} 1 & & & 0 \\ -a_1 & \ddots & & \\ \vdots & & \ddots & \\ -a_p & \dots & -a_1 & 1 \end{bmatrix} \begin{bmatrix} r_{YY}(0) \\ r_{YY}(1) \\ \vdots \\ r_{YY}(p) \end{bmatrix} + \begin{bmatrix} 0 & -a_1 & \dots & -a_p \\ \vdots & -a_2 & & 0 \\ \vdots & -a_p & 0 & \vdots \\ 0 & 0 & \dots & 0 \end{bmatrix} \begin{bmatrix} \overline{r_{YY}(0)} \\ \overline{r_{YY}(1)} \\ \vdots \\ \overline{r_{YY}(p)} \end{bmatrix} = \begin{bmatrix} \sigma_Z^2 \\ 0 \\ \vdots \\ 0 \end{bmatrix}$$

or in matrix form

$$A_1 r_{YY} + A_2 \overline{r_{YY}} = \begin{bmatrix} \sigma_Z^2 \\ 0 \end{bmatrix}$$

which can be solved for  $\{r_{YY}(k)\}_{k=0}^p$ . Then find  $r_{YY}(k)$  for  $k > p$  from equation

$$r_{YY}(k) - \sum_{i=0}^p a_i r_{YY}(k-i) = 0$$

and  $r_{YY}(k)$  for  $k < 0$  from  $\overline{r_{YY}(-k)}$ . Finally,

$$r_{XX}(k) = \sum_{i=0}^q \sum_{j=0}^q r_{YY}(k+j-i) b_i \bar{b}_j.$$